

Executing SQL Queries



Hands-on Lab : Executing SQL Queries

Estimated time needed: 30 minutes

In this lab you will be using phpMyAdmin, which is a free tool embedded in this lab environment to work on MySQL.

Objectives

After completing this lab you will be able to:

- Create a Database
- Create and load tables using csv files
- Execute SQL queries

Software used in this lab

In this lab, you will use [MySQL](#). MySQL is a Relational Database Management System (RDBMS) designed to efficiently store, manipulate, and retrieve data.



To complete this lab you will utilize MySQL relational database service available as part of IBM Skills Network Labs (SN Labs) Cloud IDE. SN Labs is a virtual lab environment used in this course.

Prework - Create and populate database

TASK A: Create a Database

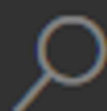
1. Start the MySQL service session using the Open MySQL Page in IDE button.

Open MySQL Page in IDE

To start the MySQL, click start.



MySQL x



MySQL

INACTIVE

8.0.22 | 5.0.4 | 2.0.2

Connect to MySQL and phpMyAdmin directly in your Skills Network Lab

Start

Stop

Summary

Connection Information

Details

Get started with MySQL in a faster, easier way. To launch your database



2. Once MySQL has started, click on phpMyAdmin button to open phpMyAdmin in the same window.

FileEditSelectionViewGoRunTerminalHelp

SKIL...MySQL x

> DATABASES

> BIG DATA

> CLOUD

> OTHER

Launch A...

MySQL

ACTIVE

v8.0.22 | v5.0.4 | v14.14

Connect to MySQL and phpMyAdmin directly in your Skills Network Labs environment.

Stop

Summary

Connection Information

Details

Your database and phpMyAdmin server are now ready to use and available with the following login credentials. For more details on how to navigate MySQL, please check out the Details section.

Username:

lavanyas

Password:

NzgyNS1sYXZhbnlh

You can manage MySQL via:

phpMyAdmin

3. You will see the phpMyAdmin GUI tool.



Recent

Favorites

- New
- + information_schema
- + mysql
- + performance_schema
- + sakila
- + sys

← Server: mysql:3306



Databases



SQL



Status



User accounts



Export



Import

General settings



Server connection collation: ?

utf8mb4_unicode_ci



More settings

Appearance settings



Language ?

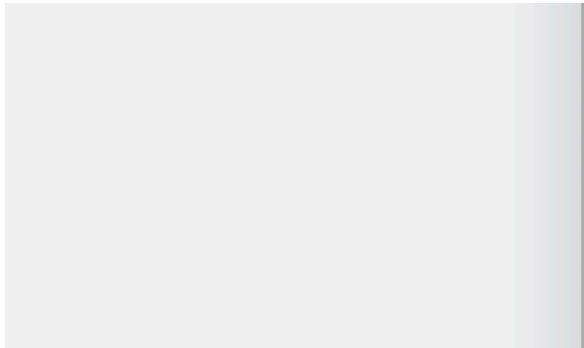
English



Theme:

pmahomme





4. In the tree-view, click **New** to create a new empty database. Then enter **Mysql_Learners** as the name of the database and select **utf8_general_ci** and click **Create**.

UTF-8 is the most commonly used character encoding for content or data.



Databases

 **Create database** 

	Database	Collation	Master replication	Action
☐	information_schema	utf8_general_ci	✓ Replicated	 Check privileges
☐	mysql	utf8mb4_0900_ai_ci	✓ Replicated	 Check privileges
☐	performance_schema	utf8mb4_0900_ai_ci	✓ Replicated	 Check privileges
☐	sys	utf8mb4_0900_ai_ci	✓ Replicated	 Check privileges
Total: 4				

TASK B: Create and load tables using csv files.

1. Download the 4 **csv** files below to your local computer:

- [dimdate.csv](#)
- [dimtruck.csv](#)
- [dimstation.csv](#)
- [facttrips.csv](#)

2. To load each **csv** file do the following steps.

- Select your database **Mysql_Learners** and click on **Import** tab and select the **csv** file.

phpMyAdmin

Server: mysql:3306 » Database: Mysql_Learners

Structure SQL Search Query Export **Import** Operations Privileges Routines Events Triggers Designer

Importing into the database "Mysql_Learners"

File to import:

File may be compressed (gzip, bzip2, zip) or uncompressed.
A compressed file's name must end in **[format].[compression]**. Example: **.sql.zip**

Browse your computer: **Choose File** DimDate.csv (Max: 2,048KiB)

You may also drag and drop a file on any page.

Character set of the file: utf-8

Partial import:

☒ Allow the interruption of an import in case the script detects it is close to the PHP timeout limit. (This might be a good way to import large files, however it can break transactions.)

Skip this number of queries (for SQL) starting from the first one: 0

Other options:

☒ Enable foreign key checks

Format:

CSV

Console file contains multiple tables, they will be combined into one.

- Then scroll down and check the box as shown below and click on **Go** to load the csv file.

phpMyAdmin

Server: mysql:3306 » Database: Mysql_Learners

Structure SQL Search Query Export Import Operations Privileges Routines Events Triggers Designer

CSV

Note: If the file contains multiple tables, they will be combined into one.

Format-specific options:

☐ Update data when duplicate keys found on import (add ON DUPLICATE KEY UPDATE)

Columns separated with: ,

Columns enclosed with: "

Columns escaped with: \"

Lines terminated with: auto

Name of the new table (optional):

Import these many number of rows (optional):

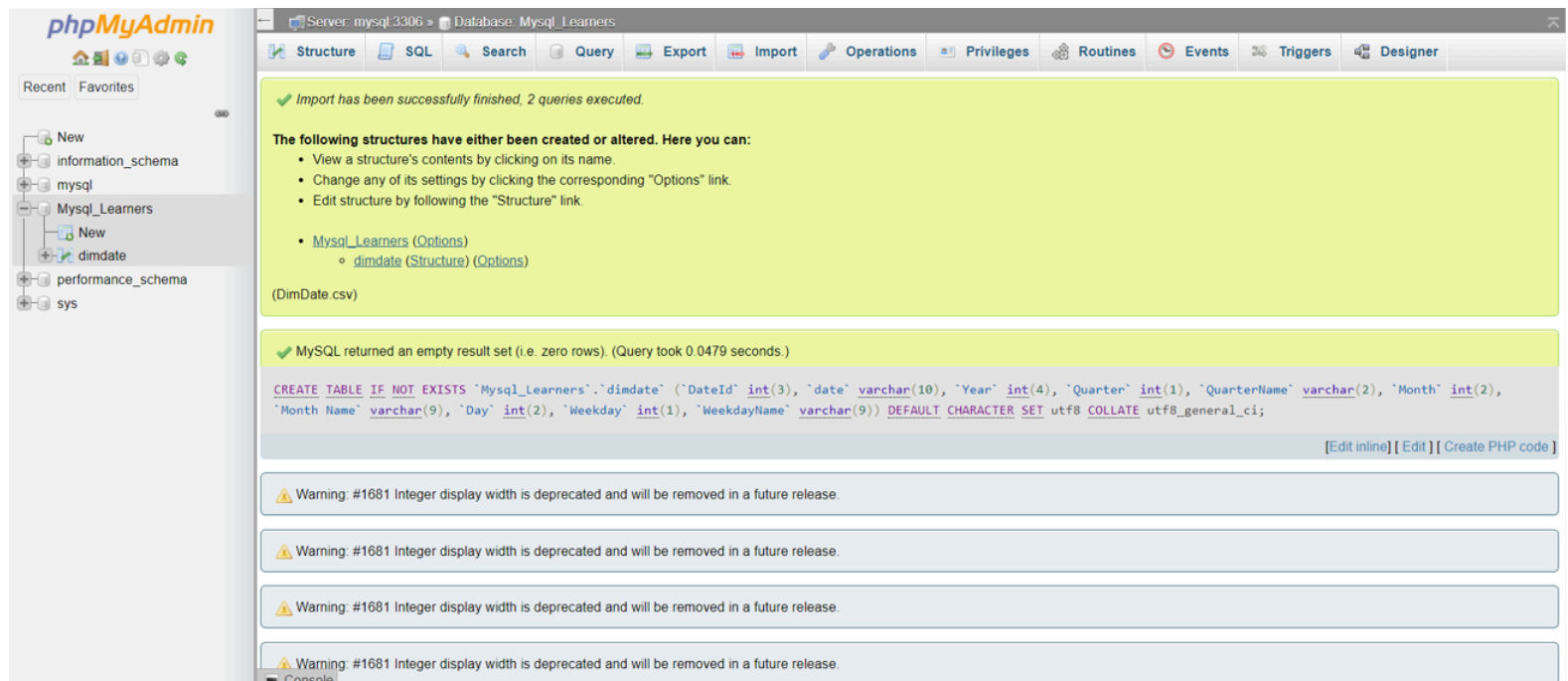
☒ The first line of the file contains the table column names (if this is unchecked, the first line will become part of the data)

☐ Do not abort on INSERT error

Console

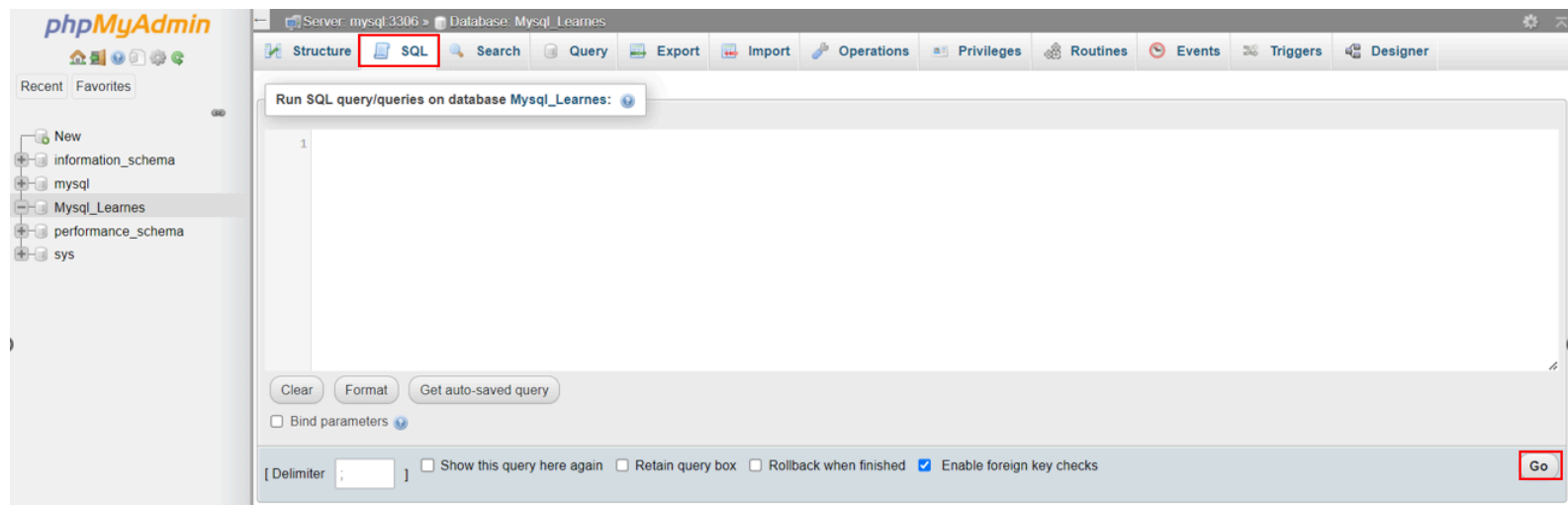
Go

- Once the tables are loaded , you will get a message that the records are inserted successfully.



Further You can import all the other csv files in the same way.

3. To run the SQL queries you need to copy the given codes and paste it to the text area of the SQL page and click on **Go**.



Execute SQL Queries

Exercise 1: List all stations in an alphabetical order. Output should contain StationId, StationName.

- ▶ Solution Syntax
- ▶ Output

Exercise 2: List all trips that collected waste > 40. Output should contain TripId, Waste.

- ▶ Solution Syntax
- ▶ Output

Exercise 3: List average waste collected for each date. Output should contain DateId, average waste.

- ▶ Solution Syntax
- ▶ Output

Exercise 4: List truck Names with their count. Output should contain TruckName, count

- ▶ Solution Syntax
- ▶ Output

Exercise 5: List City with total waste collected. Output should contain CityName, total_Waste

- ▶ Solution Syntax
- ▶ Output

Exercise 6: List minimum waste collected per quarter in 2019. Output should contain QuarterName, minimum waste.

- ▶ Solution Syntax



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- ▶ Output

Exercise 7: List maximum waste collected in Q1 in Sao Paulo. Output should contain QuarterName, City, maximum Waste.

- ▶ Solution Syntax
- ▶ Output

Exercise 8: List the days of the week results in the highest average waste collected by Volvo trucks. Output should contain WeekDayName, TruckName, avg_Waste.

- ▶ Solution Syntax
- ▶ Output

Exercise 9: List the dates when each city collected its maximum Waste. Output should contain city, date, maximum Waste.

- ▶ Solution Syntax
- ▶ Output

Congratulations! You have completed this lab successfully.

Authors

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